NYPD Shooting Incident Data Report

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Load Required Packages

```r
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Data Import

Data obtained from **https://data.cityofnewyork.us/api/views/833y-fsy8/rows.csv?accessType=DOWNLOAD**

```r
nypd_csv_path <- "NYPD_Shooting_Incident_Data__Historic_.csv"
nypd_data <- read_csv(nypd_csv_path, skip_empty_rows = TRUE)
```

```
## Rows: 29744 Columns: 21
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (12): OCCUR_DATE, BORO, LOC_OF_OCCUR_DESC, LOC_CLASSFCTN_DESC, LOCATION...
## dbl   (5): INCIDENT_KEY, PRECINCT, JURISDICTION_CODE, Latitude, Longitude
## num   (2): X_COORD_CD, Y_COORD_CD
## lgl   (1): STATISTICAL_MURDER_FLAG
## time  (1): OCCUR_TIME
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(nypd_data)
```

```
## # A tibble: 6 x 21
##   INCIDENT_KEY OCCUR_DATE OCCUR_TIME BORO     LOC_OF_OCCUR_DESC PRECINCT
##          <dbl> <chr>      <time>     <chr>    <chr>                <dbl>
## 1    231974218 08/09/2021 01:06      BRONX    <NA>                    40
## 2    177934247 04/07/2018 19:48      BROOKLYN <NA>                    79
## 3    255028563 12/02/2022 22:57      BRONX    OUTSIDE                 47
## 4     25384540 11/19/2006 01:50      BROOKLYN <NA>                    66
## 5     72616285 05/09/2010 01:58      BRONX    <NA>                    46
## 6     85875439 07/22/2012 21:35      BRONX    <NA>                    42
## # i 15 more variables: JURISDICTION_CODE <dbl>, LOC_CLASSFCTN_DESC <chr>,
## #   LOCATION_DESC <chr>, STATISTICAL_MURDER_FLAG <lgl>, PERP_AGE_GROUP <chr>,
## #   PERP_SEX <chr>, PERP_RACE <chr>, VIC_AGE_GROUP <chr>, VIC_SEX <chr>,
## #   VIC_RACE <chr>, X_COORD_CD <dbl>, Y_COORD_CD <dbl>, Latitude <dbl>,
## #   Longitude <dbl>, Lon_Lat <chr>
```

```r
summary(nypd_data)
```

```
##   INCIDENT_KEY        OCCUR_DATE          OCCUR_TIME
##  Min.   :  9953245   Length:29744       Min.   :00:00:00.000000
##  1st Qu.: 67321140   Class :character   1st Qu.:03:30:45.000000
##  Median :109291972   Mode  :character   Median :15:15:00.000000
##  Mean   :133850951                      Mean   :12:46:10.874798
##  3rd Qu.:214741917                      3rd Qu.:20:44:00.000000
##  Max.   :299462478                      Max.   :23:59:00.000000
##
##      BORO           LOC_OF_OCCUR_DESC     PRECINCT      JURISDICTION_CODE
##  Length:29744       Length:29744       Min.   :  1.00   Min.   :0.0000
##  Class :character   Class :character   1st Qu.: 44.00   1st Qu.:0.0000
##  Mode  :character   Mode  :character   Median : 67.00   Median :0.0000
##                                        Mean   : 65.23   Mean   :0.3181
##                                        3rd Qu.: 81.00   3rd Qu.:0.0000
##                                        Max.   :123.00   Max.   :2.0000
##                                                         NA's   :2
##  LOC_CLASSFCTN_DESC LOCATION_DESC      STATISTICAL_MURDER_FLAG
##  Length:29744       Length:29744       Mode :logical
##  Class :character   Class :character   FALSE:23979
##  Mode  :character   Mode  :character   TRUE :5765
##
##
##
##
##  PERP_AGE_GROUP       PERP_SEX          PERP_RACE         VIC_AGE_GROUP
##  Length:29744       Length:29744       Length:29744       Length:29744
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##    VIC_SEX            VIC_RACE           X_COORD_CD        Y_COORD_CD
```

```
##  Length:29744       Length:29744       Min.   : 914928   Min.   :125757
##  Class :character   Class :character   1st Qu.:1000094   1st Qu.:183042
##  Mode  :character   Mode  :character   Median :1007826   Median :195506
##                                        Mean   :1009442   Mean   :208722
##                                        3rd Qu.:1016739   3rd Qu.:239980
##                                        Max.   :1066815   Max.   :271128
##
##     Latitude       Longitude        Lon_Lat
##  Min.   :40.51   Min.   :-74.25   Length:29744
##  1st Qu.:40.67   1st Qu.:-73.94   Class :character
##  Median :40.70   Median :-73.91   Mode  :character
##  Mean   :40.74   Mean   :-73.91
##  3rd Qu.:40.83   3rd Qu.:-73.88
##  Max.   :40.91   Max.   :-73.70
##  NA's   :97      NA's   :97
```

```r
unique(nypd_data$VIC_RACE)
```

```
## [1] "BLACK"                          "WHITE HISPANIC"
## [3] "BLACK HISPANIC"                 "ASIAN / PACIFIC ISLANDER"
## [5] "WHITE"                          "UNKNOWN"
## [7] "AMERICAN INDIAN/ALASKAN NATIVE"
```

Clean Data

```r
# Get the number of PERP on VIC crime
nypd_data_clean <- nypd_data %>%
  filter(!is.na(PERP_RACE)) %>%
  filter(!is.na(VIC_RACE)) %>%
  filter(!(PERP_RACE %in% c('(null)', 'UNKNOWN'))) %>%
  filter(!(VIC_RACE %in% c('(null)', 'UNKNOWN'))) %>%
  mutate(PERP_ON_VIC_RACES = paste0(PERP_RACE, '>', VIC_RACE))

nypd_data_clean
```

```
## # A tibble: 16,922 x 22
##    INCIDENT_KEY OCCUR_DATE OCCUR_TIME BORO      LOC_OF_OCCUR_DESC PRECINCT
##           <dbl> <chr>      <time>     <chr>     <chr>                <dbl>
##  1    177934247 04/07/2018 19:48      BROOKLYN  <NA>                    79
##  2     72616285 05/09/2010 01:58      BRONX     <NA>                    46
##  3     85875439 07/22/2012 21:35      BRONX     <NA>                    42
##  4    142324890 04/21/2015 15:36      BROOKLYN  <NA>                    75
##  5    152868707 05/07/2016 15:23      BROOKLYN  <NA>                    69
##  6     25036119 10/30/2006 00:20      BRONX     <NA>                    40
##  7     69758511 01/10/2010 02:08      BRONX     <NA>                    42
##  8     61459877 05/06/2009 22:20      BROOKLYN  <NA>                    71
##  9     79489506 06/24/2011 04:36      BRONX     <NA>                    47
## 10     38936881 02/05/2008 20:30      MANHATTAN <NA>                    34
## # i 16,912 more rows
## # i 16 more variables: JURISDICTION_CODE <dbl>, LOC_CLASSFCTN_DESC <chr>,
## #   LOCATION_DESC <chr>, STATISTICAL_MURDER_FLAG <lgl>, PERP_AGE_GROUP <chr>,
```

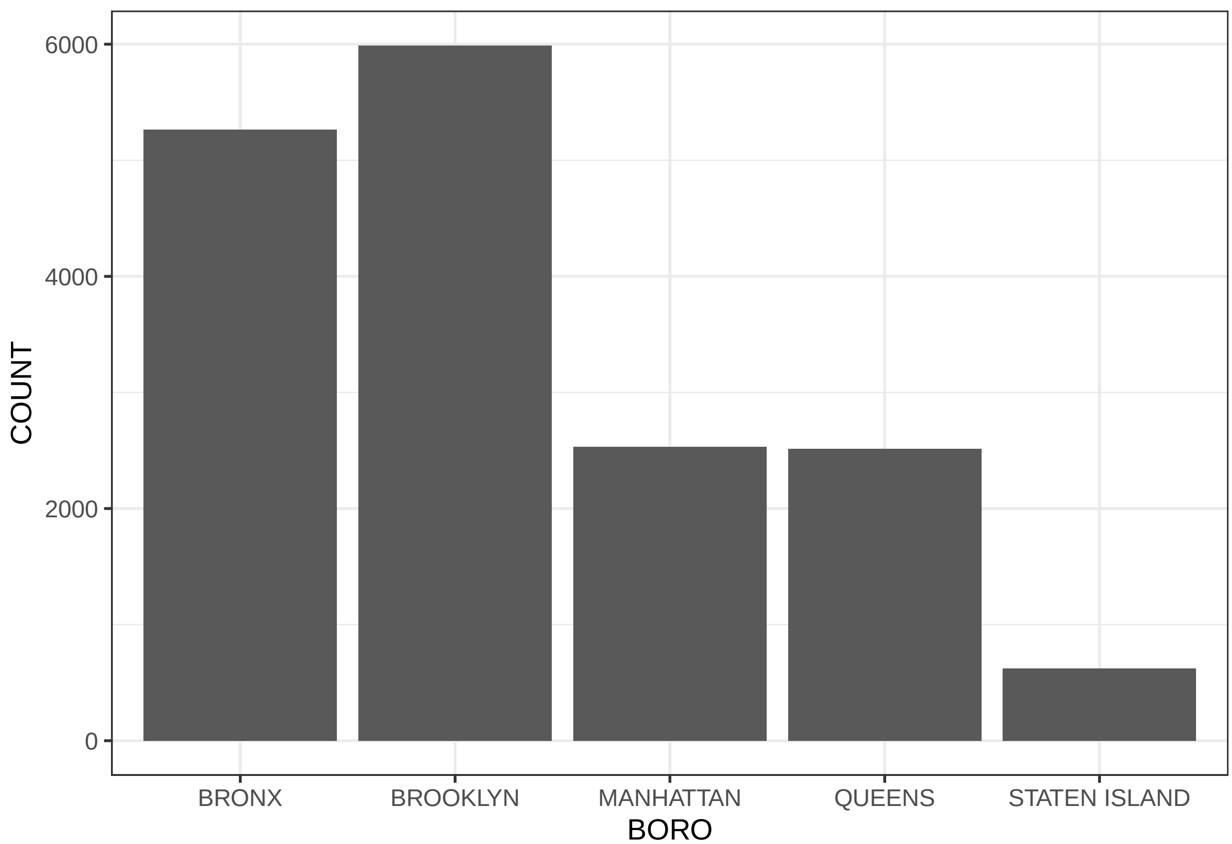
```
## #   PERP_SEX <chr>, PERP_RACE <chr>, VIC_AGE_GROUP <chr>, VIC_SEX <chr>,
## #   VIC_RACE <chr>, X_COORD_CD <dbl>, Y_COORD_CD <dbl>, Latitude <dbl>,
## #   Longitude <dbl>, Lon_Lat <chr>, PERP_ON_VIC_RACES <chr>
```

Data Analysis

Murders by Borough

```
murder_by_borough <-  nypd_data_clean %>%
  group_by(BORO) %>%
  count(BORO, name = 'COUNT')

murder_by_borough %>%
  ggplot(aes(x = BORO, y = COUNT)) +
  geom_bar(stat = 'identity') +
  theme_bw()
```

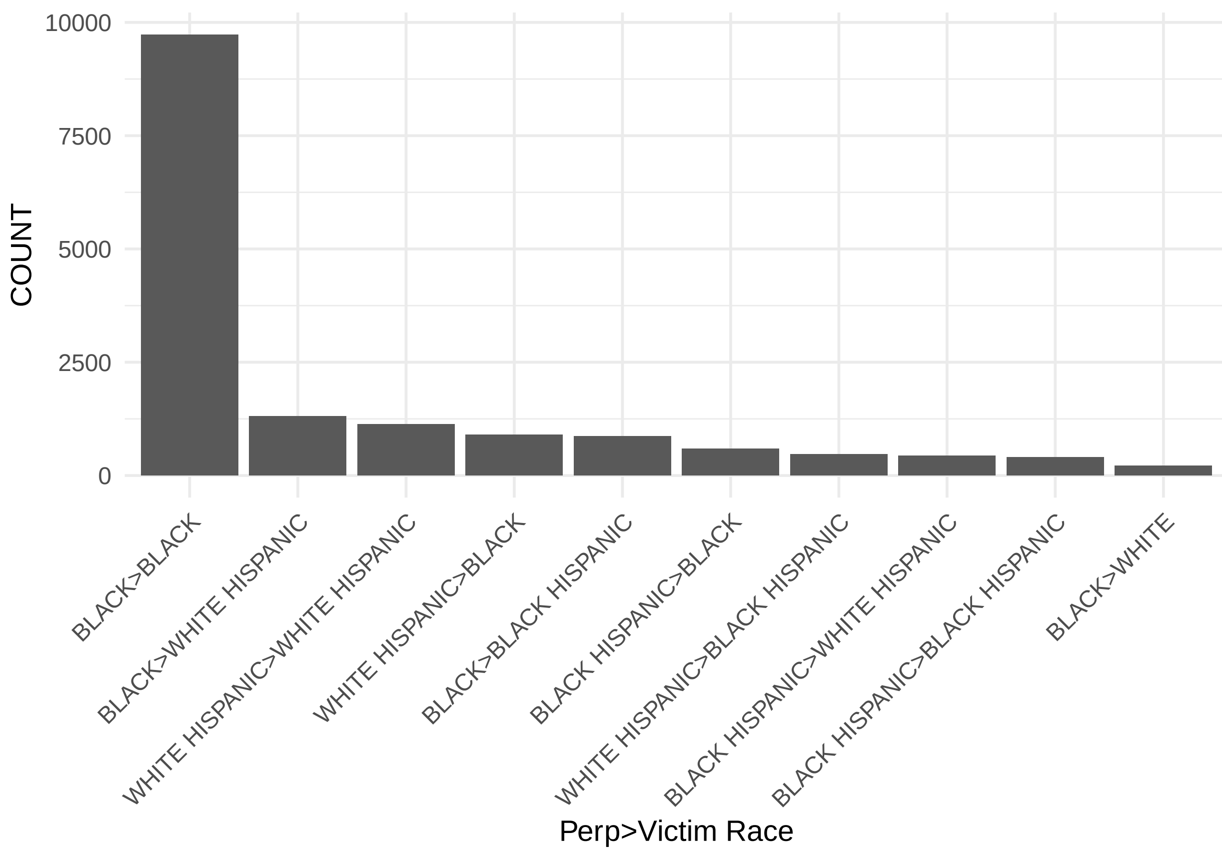


Race on Race Crime

```
race_on_race <- nypd_data_clean %>%
  count(PERP_ON_VIC_RACES, name = 'COUNT') %>%
  arrange(desc(COUNT)) %>%
```

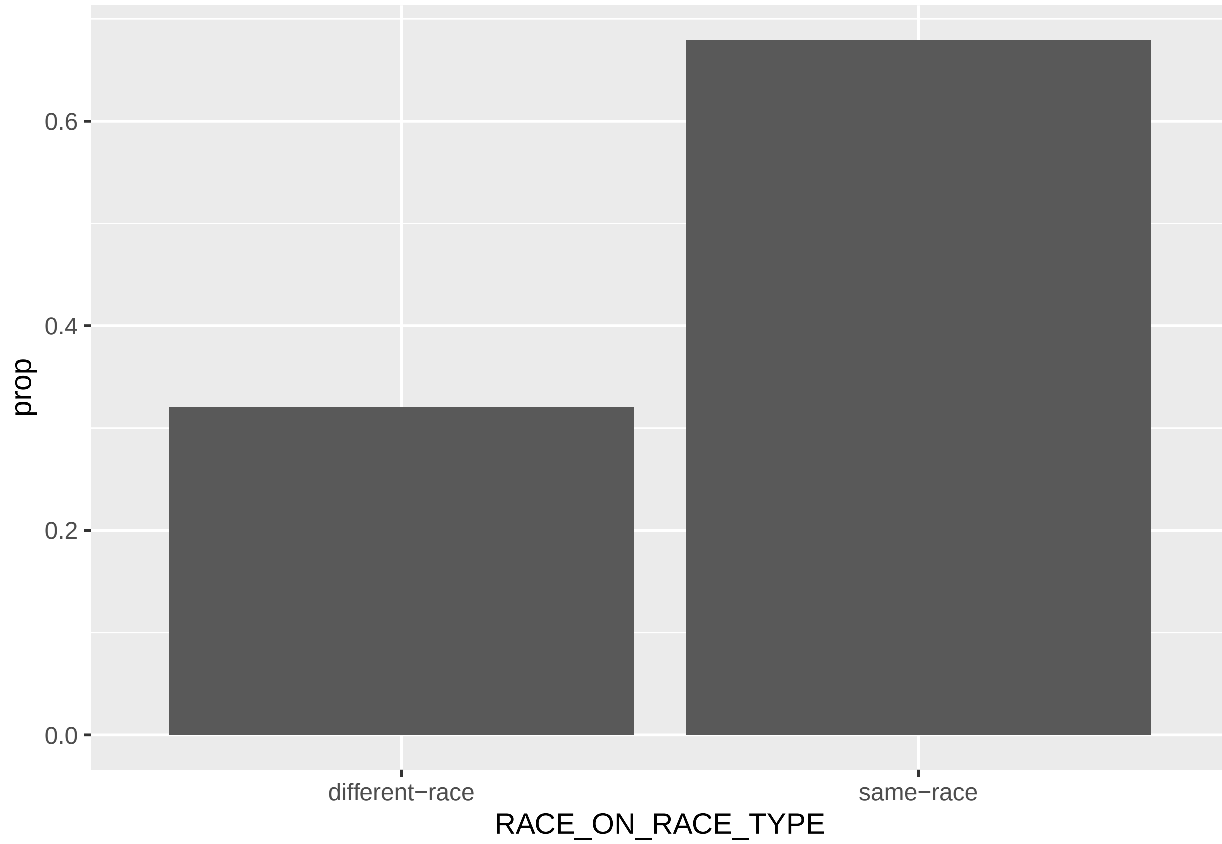
```
  head(10)

race_on_race %>%
  ggplot(aes(x = reorder(PERP_ON_VIC_RACES, -COUNT), y = COUNT)) +
  geom_bar(stat = 'identity') +
  theme_minimal() +
  theme(axis.text.x = element_text(angle=45, hjust=1)) +
  xlab('Perp>Victim Race')
```



What is the proportion on same-race versus difference-race race-on-race crime?

```
nypd_data_clean %>%
  count(PERP_ON_VIC_RACES, name = 'COUNT') %>%
  arrange(desc(COUNT)) %>%
  mutate(PERP_RACE = gsub(">.*.", "", PERP_ON_VIC_RACES)) %>%
  mutate(VICTIM_RACE = gsub(".*.>", "", PERP_ON_VIC_RACES)) %>%
  mutate(RACE_ON_RACE_TYPE = ifelse(PERP_RACE == VICTIM_RACE, "same-race", "different-race")) %>%
  group_by(RACE_ON_RACE_TYPE) %>%
  summarize(count = sum(COUNT)) %>%
  mutate(prop = count / sum(count)) %>%
  ggplot(aes(x = RACE_ON_RACE_TYPE, y = prop)) +
  geom_bar(stat='identity')
```



Data Modelling

Is the increased prevelance of same-race crime consistent across all boroughs?

```
df <- nypd_data_clean %>%
  count(PERP_ON_VIC_RACES, BORO, name = "COUNT") %>%
  mutate(
    PERP_RACE    = sub(">.*", "", PERP_ON_VIC_RACES),
    VICTIM_RACE = sub(".*>", "", PERP_ON_VIC_RACES),
    RACE_ON_RACE_TYPE = ifelse(PERP_RACE == VICTIM_RACE,
                              "same-race", "different-race")
  )

head(df)
```

```
## # A tibble: 6 x 6
##   PERP_ON_VIC_RACES           BORO  COUNT PERP_RACE VICTIM_RACE RACE_ON_RACE_TYPE
##   <chr>                       <chr> <int> <chr>     <chr>       <chr>
## 1 AMERICAN INDIAN/ALASKAN N~ BRONX     1 AMERICAN~ BLACK       different-race
## 2 AMERICAN INDIAN/ALASKAN N~ QUEE~     1 AMERICAN~ BLACK       different-race
## 3 ASIAN / PACIFIC ISLANDER>~ BRONX     3 ASIAN / ~ ASIAN / PA~ same-race
## 4 ASIAN / PACIFIC ISLANDER>~ BROO~    15 ASIAN / ~ ASIAN / PA~ same-race
## 5 ASIAN / PACIFIC ISLANDER>~ MANH~     5 ASIAN / ~ ASIAN / PA~ same-race
## 6 ASIAN / PACIFIC ISLANDER>~ QUEE~    45 ASIAN / ~ ASIAN / PA~ same-race
```

```r
model <- glm(
  COUNT ~ BORO + PERP_RACE + VICTIM_RACE + PERP_RACE:VICTIM_RACE,
  family = poisson,
  data = df
)

summary(model)
```

```
##
## Call:
## glm(formula = COUNT ~ BORO + PERP_RACE + VICTIM_RACE + PERP_RACE:VICTIM_RACE,
##     family = poisson, data = df)
##
## Coefficients: (8 not defined because of singularities)
##                                                                      Estimate
## (Intercept)                                                        -4.5932035
## BOROBROOKLYN                                                        0.1284380
## BOROMANHATTAN                                                      -0.7324770
## BOROQUEENS                                                         -0.7409088
## BOROSTATEN ISLAND                                                  -2.1338531
## PERP_RACEASIAN / PACIFIC ISLANDER                                   1.6756575
## PERP_RACEBLACK                                                      5.8128448
## PERP_RACEBLACK HISPANIC                                             4.3766486
## PERP_RACEWHITE                                                      2.3132441
## PERP_RACEWHITE HISPANIC                                             5.3341123
## VICTIM_RACEASIAN / PACIFIC ISLANDER                                 1.9420871
## VICTIM_RACEBLACK                                                    4.8965540
## VICTIM_RACEBLACK HISPANIC                                           4.2489185
## VICTIM_RACEWHITE                                                    2.7458999
## VICTIM_RACEWHITE HISPANIC                                           5.1201409
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER  4.0424137
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER                  0.8301753
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER         0.3229120
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER                  2.1085501
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER                NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK                  0.9647142
## PERP_RACEBLACK:VICTIM_RACEBLACK                                     1.8997250
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK                            0.5278740
## PERP_RACEWHITE:VICTIM_RACEBLACK                                     0.0229162
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK                                   NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC         0.1780753
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC                            0.1247033
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC                   0.7894461
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC                           -0.0006165
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC                          NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE                  1.4894011
## PERP_RACEBLACK:VICTIM_RACEWHITE                                     0.2191653
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE                           -0.0855787
## PERP_RACEWHITE:VICTIM_RACEWHITE                                     3.4908719
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE                                   NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC                NA
## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC                           -0.3268640
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC                          NA
```

```
## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC                                        NA
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC                               NA
##                                                                         Std. Error
## (Intercept)                                                              1.2254656
## BOROBROOKLYN                                                             0.0188966
## BOROMANHATTAN                                                            0.0241879
## BOROQUEENS                                                               0.0242471
## BOROSTATEN ISLAND                                                        0.0424342
## PERP_RACEASIAN / PACIFIC ISLANDER                                        0.7333326
## PERP_RACEBLACK                                                           1.3043955
## PERP_RACEBLACK HISPANIC                                                  0.7101930
## PERP_RACEWHITE                                                           0.7212812
## PERP_RACEWHITE HISPANIC                                                  0.7079394
## VICTIM_RACEASIAN / PACIFIC ISLANDER                                      1.0107500
## VICTIM_RACEBLACK                                                         1.0007236
## VICTIM_RACEBLACK HISPANIC                                                1.0012281
## VICTIM_RACEWHITE                                                         1.0049193
## VICTIM_RACEWHITE HISPANIC                                                1.0006126
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER    0.2690164
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER                       1.1078327
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER              0.2571799
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER                       0.3424832
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER                     NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK                       0.2326052
## PERP_RACEBLACK:VICTIM_RACEBLACK                                          1.0961209
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK                                 0.0775127
## PERP_RACEWHITE:VICTIM_RACEBLACK                                          0.2059107
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK                                        NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC              0.3318856
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC                                 1.0970629
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC                        0.0884359
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC                                 0.2542936
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC                               NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE                       0.3598031
## PERP_RACEBLACK:VICTIM_RACEWHITE                                          1.1020584
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE                                 0.1993661
## PERP_RACEWHITE:VICTIM_RACEWHITE                                          0.1858616
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE                                        NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC                     NA
## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC                                 1.0963201
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC                               NA
## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC                                        NA
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC                               NA
##                                                                         z value
## (Intercept)                                                              -3.748
## BOROBROOKLYN                                                              6.797
## BOROMANHATTAN                                                           -30.283
## BOROQUEENS                                                              -30.557
## BOROSTATEN ISLAND                                                       -50.286
## PERP_RACEASIAN / PACIFIC ISLANDER                                         2.285
## PERP_RACEBLACK                                                            4.456
## PERP_RACEBLACK HISPANIC                                                   6.163
## PERP_RACEWHITE                                                            3.207
## PERP_RACEWHITE HISPANIC                                                   7.535
```

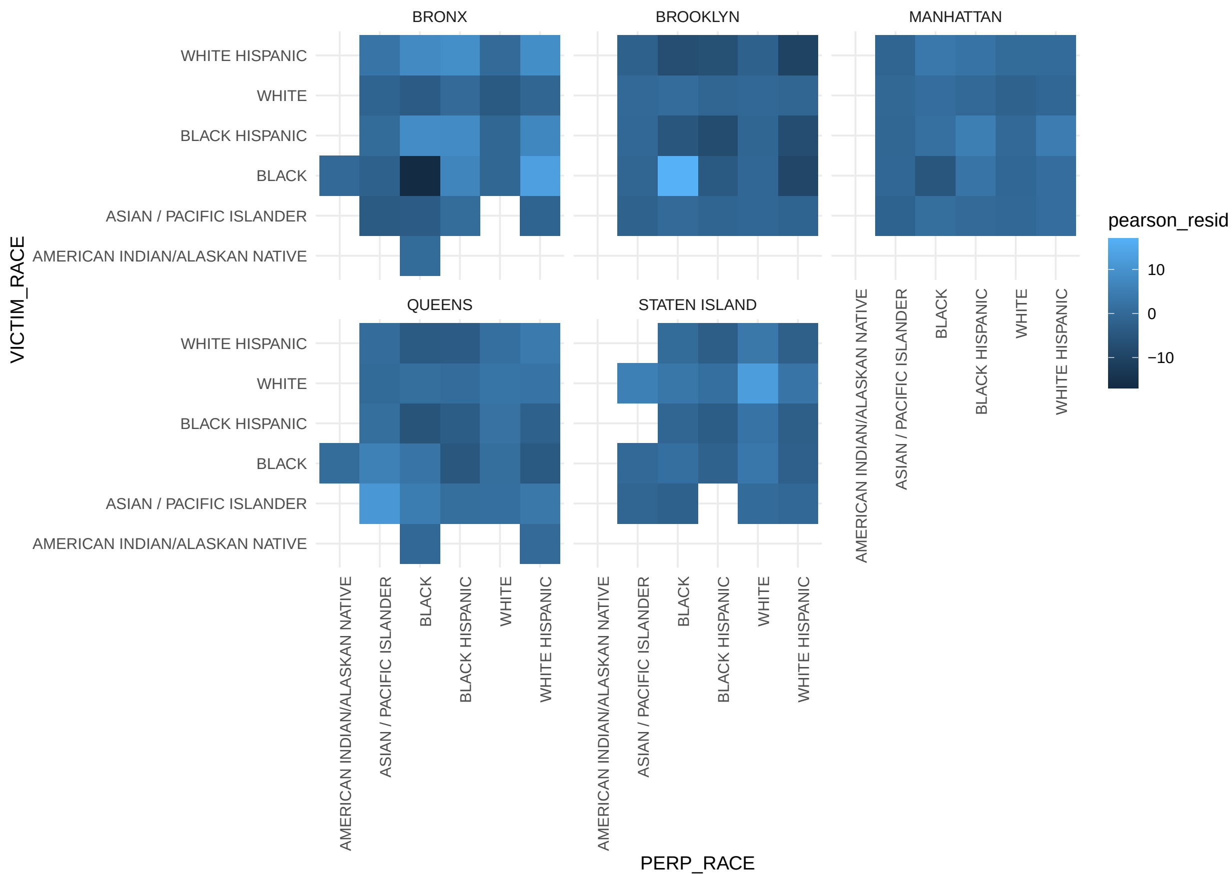
```
## VICTIM_RACEASIAN / PACIFIC ISLANDER                                 1.921
## VICTIM_RACEBLACK                                                    4.893
## VICTIM_RACEBLACK HISPANIC                                           4.244
## VICTIM_RACEWHITE                                                    2.732
## VICTIM_RACEWHITE HISPANIC                                           5.117
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER  15.027
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER                  0.749
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER         1.256
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER                  6.157
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER            NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK                  4.147
## PERP_RACEBLACK:VICTIM_RACEBLACK                                     1.733
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK                            6.810
## PERP_RACEWHITE:VICTIM_RACEBLACK                                     0.111
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK                               NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC         0.537
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC                            0.114
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC                   8.927
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC                           -0.002
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC                      NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE                  4.139
## PERP_RACEBLACK:VICTIM_RACEWHITE                                     0.199
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE                           -0.429
## PERP_RACEWHITE:VICTIM_RACEWHITE                                    18.782
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE                               NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC            NA
## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC                           -0.298
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC                      NA
## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC                               NA
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC                      NA
##                                                                  Pr(>|z|)
## (Intercept)                                                      0.000178
## BOROBROOKLYN                                                     1.07e-11
## BOROMANHATTAN                                                     < 2e-16
## BOROQUEENS                                                        < 2e-16
## BOROSTATEN ISLAND                                                 < 2e-16
## PERP_RACEASIAN / PACIFIC ISLANDER                                0.022313
## PERP_RACEBLACK                                                   8.34e-06
## PERP_RACEBLACK HISPANIC                                          7.16e-10
## PERP_RACEWHITE                                                   0.001341
## PERP_RACEWHITE HISPANIC                                          4.89e-14
## VICTIM_RACEASIAN / PACIFIC ISLANDER                              0.054677
## VICTIM_RACEBLACK                                                 9.93e-07
## VICTIM_RACEBLACK HISPANIC                                        2.20e-05
## VICTIM_RACEWHITE                                                 0.006286
## VICTIM_RACEWHITE HISPANIC                                        3.10e-07
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER  < 2e-16
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER               0.453635
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER      0.209265
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER               7.43e-10
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER            NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK               3.36e-05
## PERP_RACEBLACK:VICTIM_RACEBLACK                                  0.083072
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK                         9.75e-12
```

```
## PERP_RACEWHITE:VICTIM_RACEBLACK                                    0.911385
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK                                 NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC        0.591574
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC                           0.909499
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC                   < 2e-16
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC                           0.998066
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC                        NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE                 3.48e-05
## PERP_RACEBLACK:VICTIM_RACEWHITE                                    0.842365
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE                           0.667738
## PERP_RACEWHITE:VICTIM_RACEWHITE                                     < 2e-16
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE                                 NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC              NA
## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC                           0.765591
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC                        NA
## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC                                 NA
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC                        NA
##
## (Intercept)                                                        ***
## BOROBROOKLYN                                                       ***
## BOROMANHATTAN                                                      ***
## BOROQUEENS                                                         ***
## BOROSTATEN ISLAND                                                  ***
## PERP_RACEASIAN / PACIFIC ISLANDER                                  *
## PERP_RACEBLACK                                                     ***
## PERP_RACEBLACK HISPANIC                                            ***
## PERP_RACEWHITE                                                     **
## PERP_RACEWHITE HISPANIC                                            ***
## VICTIM_RACEASIAN / PACIFIC ISLANDER                                .
## VICTIM_RACEBLACK                                                   ***
## VICTIM_RACEBLACK HISPANIC                                          ***
## VICTIM_RACEWHITE                                                   **
## VICTIM_RACEWHITE HISPANIC                                          ***
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER ***
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER                 ***
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK                 ***
## PERP_RACEBLACK:VICTIM_RACEBLACK                                    .
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK                           ***
## PERP_RACEWHITE:VICTIM_RACEBLACK
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC                  ***
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE                 ***
## PERP_RACEBLACK:VICTIM_RACEWHITE
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE
## PERP_RACEWHITE:VICTIM_RACEWHITE                                    ***
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC
```

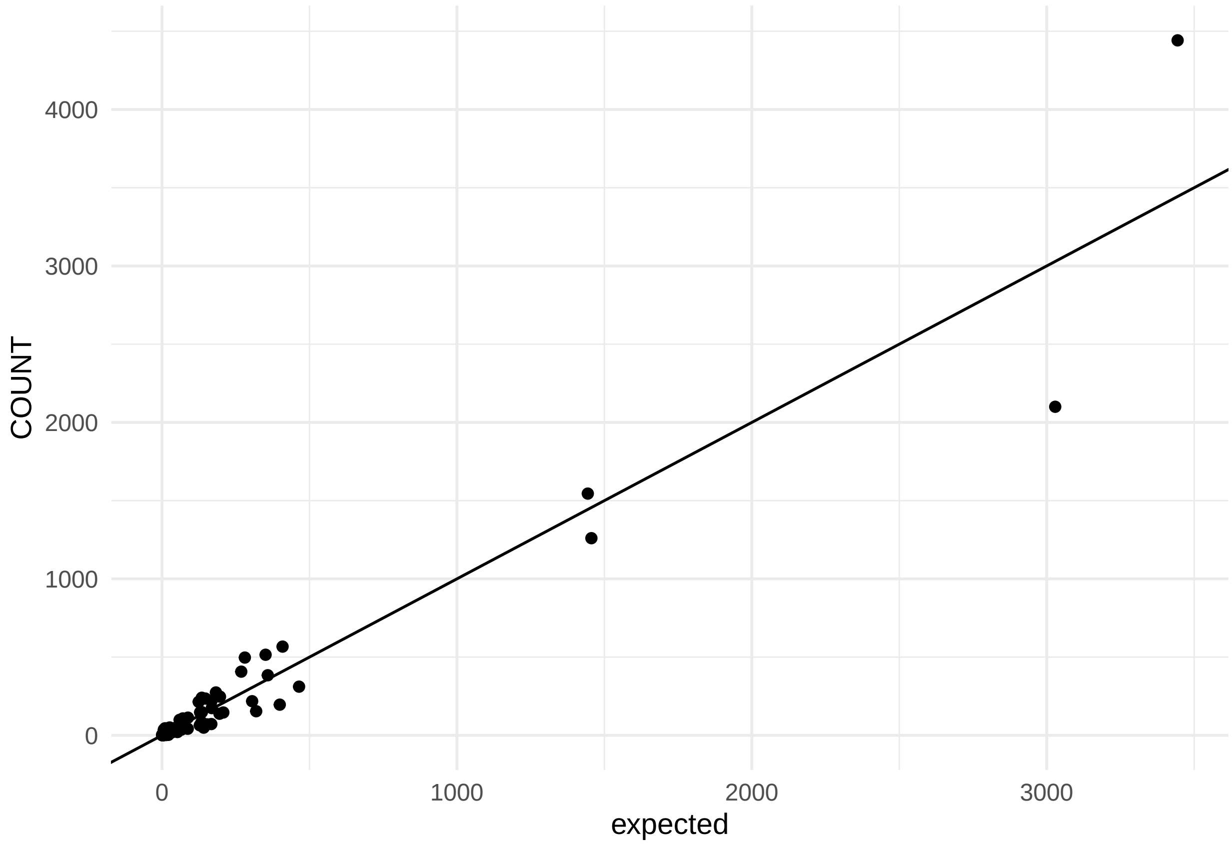
```
## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC
## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 59173.8  on 125  degrees of freedom
## Residual deviance:  2368.5  on  94  degrees of freedom
## AIC: 3042.8
##
## Number of Fisher Scoring iterations: 5
```

```
df$expected <- predict(model, type = "response")
df$pearson_resid <- residuals(model, type = "pearson")

ggplot(df, aes(PERP_RACE, VICTIM_RACE, fill = pearson_resid)) +
  geom_tile() +
  facet_wrap(~ BORO) +
  theme_minimal() +
  coord_equal() +
  theme(axis.text.x.bottom = element_text(angle=90, hjust=1))
```




```r
ggplot(df, aes(expected, COUNT)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0) +
  theme_minimal()
```



Looks fairly 1-to-1 linear, which makes sense since we're comparing against the data we have trained again. To test this appropriately most likely would need to do cross-validation.

Brief comment on Bias

The analysis may be biased towards certain demographics which have a larger proportion of crimes and skew the definition of same-race crime. For example, certain ethnicity in New York may be less prevalent and therefore less likely to conduct same-race crimes, or more likely to conduct different-race crime. It's important to consider this when conducting analysis like this.